

A Creating and adding data to the value creation catalogue

This appendix contains a guide explaining how to create new data for the value creation catalogue as well as how to add it into the catalogue. This appendix is intended to function as a stand-alone guide, meaning that following the guide should not require reading the accompanying thesis first. The only prerequisite to following the guide is that the reader has the value creation catalogue and its documentation downloaded locally.

A.1 Creating a new EoL scenario for the catalogue

A.1.1 Declared unit & mass factor

Before modelling the LCA calculation for a new EoL scenario, the declared unit and the mass factor must be determined. The declared unit refers to whichever unit is used to quantify the material, and the mass factor is the weight per declared unit.

If you are modelling a new EoL scenario for a material that already exists in the catalogue, both the declared unit and mass factor will already be defined within the catalogue (see example in Figure A.1).

Material	Unit	Kg/Unit
Construction Wood	 m ³	488
	ID#	Scenario
	CW IN 01	Wood extracted from I (diesel) and transport

Figure A.1: Declared unit (left) and mass factor (right) of construction wood.

If you are modelling an EoL scenario for a material that is not yet added to the catalogue, you will need to define the declared unit and mass factor yourself. It is recommended that you find a Danish industry EPD representing the material, if available, and use the declared unit and mass factor defined in the EPD. If no industry EPD is available, another option is to find the material in BR18, appendix 2, table 7 [34] and use the declared unit and mass factor defined there.

A.1.2 Data collection

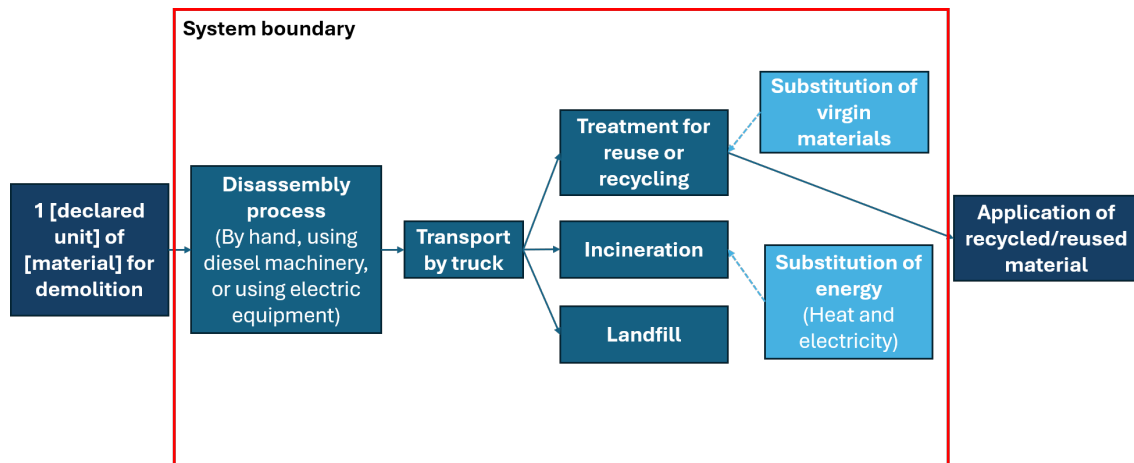


Figure A.2: Visual representation of the system boundary for LCA calculations conducted in the study.

Considering the system boundary, shown in Figure A.2, and the type of EoL treatment being modelled, the following data must be collected:

Landfill

1. **Disassembly method:** How is the material being removed from the building?
2. **Transport:** How far is the material transported, and what method of transport is used?
3. **Treatment:** If the material receives further treatment prior to landfill, what treatment does it receive?

Incineration

1. **Disassembly method:** How is the material being removed from the building?
2. **Transport:** How far is the material transported, and what method of transport is used?
3. **Treatment:** If the material receives further treatment prior to incineration, what treatment does it receive?
4. **Energy recovery:** How much electricity and heat is recovered from incinerating the material?

Recycling

1. **Disassembly method:** How is the material being removed from the building?
2. **Transport:** How far is the material transported, and what method of transport is used?
3. **Treatment:** What treatment, if any, does the material need before being recycled?
4. **Recyclability:** How much of the product is recycled, and what happens to the remaining material?
5. **Avoided product:** What product does the recycled material replace?

Reuse

1. **Disassembly method:** How is the material being removed from the building?
2. **Transport:** How far is the material transported, and what method of transport is used?
3. **Treatment:** What treatment, if any, does the material need before it can be reused?
4. **Reusability:** How much of the product is reused, and what happens to the remaining material?
5. **Avoided product:** What product does the reused material replace?

The catalogue uses generalised assumptions for the energy use of the disassembly process depending on whether it is done by hand, using diesel machinery, or electric tools. These assumptions can be seen in Table A.1.

Table A.1: Energy use/consumption assumed in the study depending on disassembly method.

Disassembly method	Energy use / consumption
Manual (by hand)	none
Diesel machinery	1.4 mL diesel/kg material
Electric tools	989 J electricity/kg material

The catalogue also uses generalised assumptions for the transportation process depending on whether the material is transported to final disposal (landfill/incineration) or a specialised facility (recycling/reuse), and whether the specialised material is in Denmark or not. All transportation is assumed to happen via truck. These assumptions can be seen in Table A.2.

Table A.2: Transport distance and truck size assumed in the study depending where the material is transported to.

Transport location	Distance	Truck size
Final disposal	50 km	16-32 ton
Specialised facility (Denmark)	100 km	16-32 ton
Specialised facility (International)	1000 km	>32 ton

Collection of data for the remaining processes should follow the hierarchy described in Figure A.3, prioritising gathering data from as high on the hierarchy as possible.

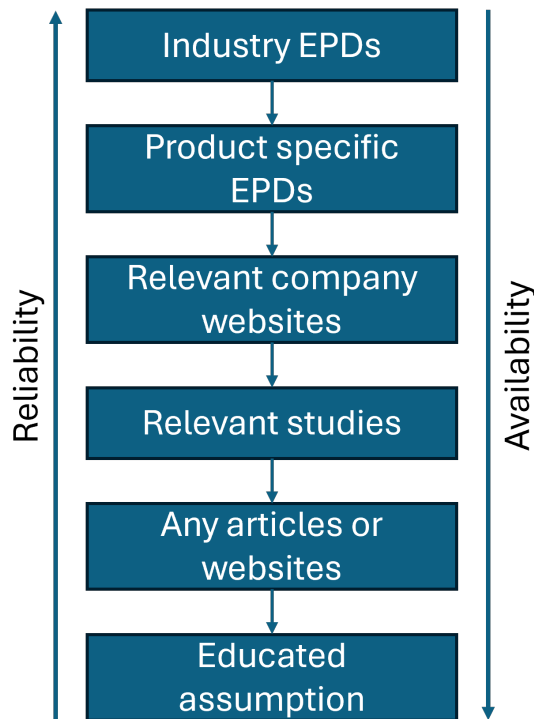


Figure A.3: Visual representation of the hierarchy used for data collection for the catalogue.

A.1.3 LCA calculation

The functional unit (FU) for LCA calculations in the catalogue is defined as:

“1 [declared unit] of [material] removed from the building for end-of-life treatment”

The LCA calculations must include the processes within the system boundary shown in Figure A.2 required for fulfilling 1 FU.

A.1.4 Environmental data

The LCA calculation must use environmental data from the EcolInvent database. All EoL scenarios created prior to the 9th of August 2026 were calculated using environmental data from the 3.12 version of the EcolInvent database. As future versions of the EcolInvent database are released, these may be used instead.

Since the EoL scenarios are meant to represent EoL treatment of materials in the Danish building industry, the geographical representativeness should be considered when choosing environmental data. Ideally, environmental data with the geography code 'DK' (Denmark) should be used (excluding treatment outside of Denmark), however, this is not available for many of the relevant processes. Usually, the best approximation will have the geography code 'RER' or 'Europe without Switzerland', and in some cases even 'RoW' (Rest of World) or 'GLO' (Global).

For transportation, calculations must use the following EcolInvent processes:

Truck size: 16-32 ton

Transport, freight, lorry 16-32 metric ton, euro6 RER| market for transport, freight, lorry 16-32

metric ton, EURO6 | Cut-off, S

Truck size: >32 ton

Transport, freight, lorry, >32 metric ton, diesel, EURO 6 RER| market for transport, freight, lorry, >32 metric ton, diesel, EURO 6 | Cut-off, S

For disassembly, calculations must use the following Ecolnvent processes:

Diesel consumption (assume diesel density of 0.845 kg/L with an energy density of 42.7 MJ/kg)

Diesel, burned in building machine GLO| market for diesel, burned in building machine | Cut-off, S

Electricity use

Electricity, low voltage DK| market for | Cut-off, S

Note that the names of processes may differ in future version of the Ecolnvent database.

A.1.5 Impact Assessment Method

LCA calculations must be performed using the ReCiPe2016 Midpoint (H) impact assessment method. The catalogue will need the LCA results for the following impact categories:

1. Global warming
2. Land use
3. Mineral resource scarcity
4. Fossil resource scarcity

A.1.6 Calculation software

There is no direct requirement of which software to use for the calculations. All EoL scenarios created prior to the 9th of August 2026 were modelled and calculated using SimaPro desktop edition (version 10.4.0.0), but any software is permitted as long as it lives up to the calculation requirements.

A.1.7 Documentation

Once the LCA calculation has been performed, it must be documented before it can be added into the catalogue. In the 'Templates' folder, find the file 'Documentation_Template.xlsx' and make a copy of it. The copy should be renamed to the ID-number of the scenario.

Determining the ID-number

Every EoL scenario in the catalogue has to be assigned an ID-number in the format 'XX_YY_ZZ'.

The first two symbols (XX) should be letters representing the material. Ensure that no two materials are given the same signifier, potentially by adding a third letter if needed.

The second two symbols (YY) should be letters representing the EoL treatment.

- Landfill: LF

- Incineration: **IN**
- Recycling: **RC**
- Reuse: **RU**

The third two symbols (ZZ) should be a number in case there is more than one scenario for the material with the same type of EoL treatment.

For example, the incineration scenario for construction wood has been assigned the ID-number 'CW_IN_01'. The corresponding documentation file, found in the 'Documentation' folder, is then 'CW_IN_01.xlsx'.

Fill in the documentation template

Open the copied template and fill in the information of the EoL scenario and the results of the LCA calculation.

In the first row, enter the ID-number of the scenario on the left side of the page and the date when the calculation was performed on the right (see Figure A.4). Also name the sheet itself to match the ID-number.

XX_YY_ZZ - Life Cycle Inventory	Page 1 of 1	dd-mm-yyyy
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CW_IN_01 - Life Cycle Inventory	Page 1 of 1	18-06-2026
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Figure A.4: How to fill in the top row of the documentation sheet using the incineration scenario for construction wood as an example.

Next, fill in the name of the material, declared unit, mass factor, and the scenario name in their respective fields (see Figure A.5).

Material	[material name]	→	Material	Construction Wood
Declared unit	1 [declared unit]		Declared unit	1 m ³
Mass factor	[mass factor] kg/[declared unit]		Mass factor	488 kg/m ³
Scenario	[scenario name]		Scenario	Incineration

Figure A.5: How to fill in the properties of the scenario in the documentation sheet using the incineration scenario for construction wood as an example.

Then, write a description of the scenario including key assumptions (see Figure ??).

Scenario description
[write a description of the scenario including key assumptions]

Scenario description
1 m³ of construction wood is extracted from the building diesel-driven machinery. Assumed diesel use of 1.4 L/t material based on a report from the Danish Environmental Protection Agency (2022) [1]. The construction wood is transported 50 km by truck to a municipal incineration plant where it is incinerated for energy recovery. Heat recovery of 1430 MJ/m³ construction wood and electricity recovery of 712 MJ/m³ construction wood from Danish industry EPD for construction wood [2].

Figure A.6: How to fill in the scenario description in the documentation sheet using the incineration scenario for construction wood as an example.

Then, fill out the LCI with a short name describing each process, the EcolInvent data used to represent the process, and how much of each process is included per FU (see Figure A.7). You can add or remove rows as necessary.

Life Cycle Inventory			
Process	EcolInvent process	Amount	Unit
[short name describing process]	[name of EcolInvent process]	[amount]	[unit]
[short name describing process]	[name of EcolInvent process]	[amount]	[unit]
[short name describing process]	[name of EcolInvent process]	[amount]	[unit]

Life Cycle Inventory			
Process	EcolInvent process	Amount	Unit
Disassembly	Diesel, burned in building machine {GLO} market for diesel, burned in building machine Cut-off, S	24.7	MJ
Transportation	Transport, freight, lorry 16-32 metric ton, euro6 {RER} market for transport, freight, lorry 16-32 metric ton, EURO6 Cut-off, S	24.4	tkm
Incineration	Waste wood, untreated {GLO} treatment of waste wood, untreated, municipal incineration Cut-off, S	488	kg
Recovered heat	Heat, district or industrial, other than natural gas {Europe without Switzerland} market for heat, district or industrial, other than natural gas Cut-off, S	-1.43	GJ
Recovered electricity	Electricity, low voltage {DK} market for Cut-off, S	-712	MJ

Figure A.7: How to fill in the LCI of the scenario in the documentation sheet using the incineration scenario for construction wood as an example.

The next step is to fill in the contribution of each process to each environmental impact category as well as the total environmental impacts per FU (see Figure ??). You can add or remove rows as necessary.

Process contribution				
Process	Global Warming [kg CO ₂ eq]	Land Use [m ² a crop eq]	Mineral Resource Scarcity [kg CU eq]	Fossil Resource Scarcity [kg oil eq]
[short name describing process]	0.00E+00	0.00E+00	0.00E+00	0.00E+00
[short name describing process]	0.00E+00	0.00E+00	0.00E+00	0.00E+00
[short name describing process]	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Total of all processes	0.00E+00	0.00E+00	0.00E+00	0.00E+00



Process contribution				
Process	Global Warming [kg CO ₂ eq]	Land Use [m ² a crop eq]	Mineral Resource Scarcity [kg CU eq]	Fossil Resource Scarcity [kg oil eq]
Disassembly	2.53E+00	1.78E-02	4.08E-03	7.44E-01
Transportation	4.66E+00	2.07E-01	1.23E-02	1.47E+00
Incineration	7.70E+00	1.26E-01	6.49E-02	1.57E+00
Recovered heat	-1.04E+02	-6.10E+00	-3.38E-02	-2.27E+01
Recovered electricity	-3.53E+01	-7.98E+00	-3.07E-01	-8.71E+00
Total of all processes	-1.25E+02	-1.37E+01	-2.60E-01	-2.76E+01

Figure A.8: How to fill in the environmental impacts of the scenario in the documentation sheet using the incineration scenario for construction wood as an example.

Then, write any other relevant information in the 'Comments' field (see Figure ??). The EcolInvent database version and the impact assessment method must be included as the bare minimum as well as URLs for any references used.

Comments
EcolInvent database version: [EcolInvent version] Impact assessment method: ReCiPe2016 Midpoint (H) [other relevant comments like references, description of mass factor etc.]



Comments
EcolInvent database version: 3.12 Impact assessment method: ReCiPe2016 Midpoint (H) Mass factor determined from Danish industry EPD for construction wood MD-25104-EN [2] Diesel density assumed to be 0.845 kg/L with an energy density of 42.7 MJ/kg. [1] Miljøstyrelsen (2022): https://www2.mst.dk/Udgiv/publikationer/2022/02/978-87-7038-353-0.pdf [2] MD-25104-EN: https://www.epddanmark.dk/media/y1wfvthh/md-25104-en.pdf

Figure A.9: How to fill in the 'Comments' field in the documentation sheet using the incineration scenario for construction wood as an example.

If the sheet now takes up more than one page of space, copy the first row of the first page and insert it as the first row on the second page to indicate the start of the new page. Remember to change the page numbers to be '1 of 2' and '2 of 2' for the first and second page respectively.

Once the documentation sheet is completed, move it to the 'Documentation' folder.

A.2 Inserting data into the catalogue

It is recommended that you make a backup of the catalogue before adding any new data into it.

A.2.1 Assign a circularity level

Before adding a new EoL scenario to the catalogue, it must be assigned a circularity level. The circularity levels each represent a type of EoL treatment with a lower number corresponding to a more circular EoL scenario. The circularity levels are defined as follows:

Circularity levels

- **0** - Represents a **Reuse** scenario. This means that the material is directly reused as is without any major processing being required. (Example: bricks being cleaned and reused in a new building)
- **1** - Represents a closed loop **Recycling** scenario. This means that the material is processed to be remade into the same material again. (Example: steel being melted into new steel products)
- **2** - Represents an open loop **Recycling** scenario. This means that the material is processed to be remade into a new material. (Example: window glass being used in glass wool production)
- **3** - Represents a **Recovery** scenario. This means that the material is processed to obtain energy. (Example: construction wood incinerated to generate heat and electricity)
- **4** - Represents a **Disposal** scenario. This means that the material is disposed of in a way that generates no value. (Example: glass wool being disposed of in a landfill)

A.2.2 Inserting a new scenario into existing material

Open '*Value Creation Catalogue.xlsm*' and navigate to the sheet named '*Catalogue*'. Unhide columns C and P in the sheet. Next, navigate to the material for which you wish to add the new scenario, and expand it by clicking on the '+'-symbol on the left side of the screen.

Select the row of one of the scenarios that are already there for the material, except for the first scenario. Copy the row and insert it somewhere between the first and last EoL scenario of the material (see Figure A.10).



Figure A.10: How to insert a row for the new scenario using the construction wood as an example.

In the newly inserted row, replace the information with the information of the new scenario. From left to right, this includes ID-number, name of the scenario, a short description of the scenario, and the circularity level.

Next, enter the total results of the LCA calculation under the respective impact category. This should be done as a formula multiplying the amount (the cell to the right of the material image and to the left of the declared unit of the material) with the calculation results (see Figure A.11). It should be done this way so that the results can be scaled when added to a building scenario.

Cell ID of the 'Amount' of material

$$=C46 * -60.7$$

Result of the calculation

Figure A.11: How to enter results in the catalogue.

Then, ensure that only the row containing the last (bottom) scenario of the material has the name of the material followed by 1 in column P. Once that is done, hide columns C and P again.

A.2.3 Inserting a new material

Note that for a new material to be added to the catalogue, it must include at least two EoL scenarios.

Open 'Value Creation Catalogue.xlsm' and navigate to the sheet named 'Catalogue'. Unhide columns C, P, and Q in the sheet. Next, expand any material by clicking on the '+'-symbol on the left side of the screen.

Select all the rows of a material that is already in the catalogue. Copy the rows and insert it somewhere between two materials or at the end of the catalogue (see Figure A.12).

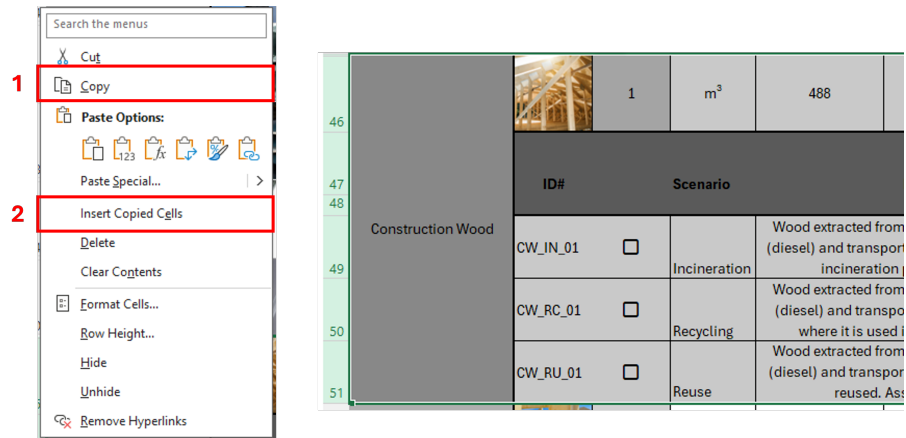


Figure A.12: How to copy (1) and insert (2) the rows for a new material.

In the newly inserted rows, replace the image and information with the image and information of the new material and the individual scenarios. See subsection A.2.2 for how to enter scenario data. You can add rows for more scenarios or delete rows to fit the number of scenarios for the new material. Note, you can replace the information in the first (top) scenario, but do not delete the row, as this will break the visual comparison graph.

Then, ensure that the top row of the material has the name of the material followed by 1 in column P, and that only the row containing the last (bottom) scenario of the material has the name of the material followed by 1 in column P. In the top row of the material, you can insert a URL for the image of the material in column Q. Once all that is done, hide columns C, P, and Q again.

The visual comparison

By default, the visual comparison graph will still refer to the material it was originally copied from. The first step to fixing this is to update the cells hidden behind the comparison graph. You can move the graph aside to see the cells.

The cell in the top left looks at the impact category chosen in the drop down, and finds the appropriate unit. The formula should look something like Figure A.13.

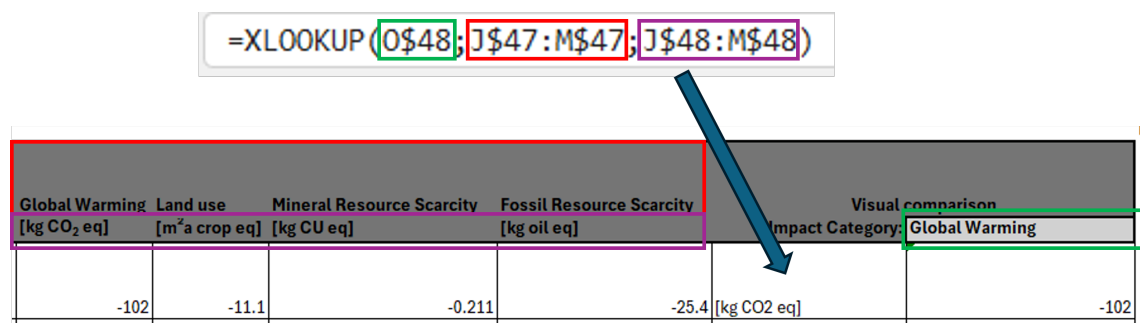


Figure A.13: Formula identifying the appropriate unit for the graph.

The cells on the right look at the impact category chosen in the drop down, and finds the appropriate

result for each scenario. The formula should look something like Figure A.14.

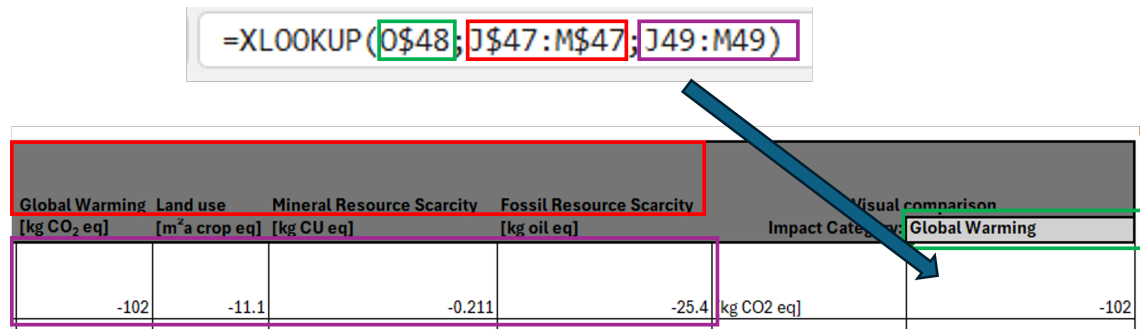


Figure A.14: Formula identifying the appropriate result for the graph.

The next step is to connect the graph to the correct data. Right-click on the graph and select 'Select Data...' (see Figure A.15).

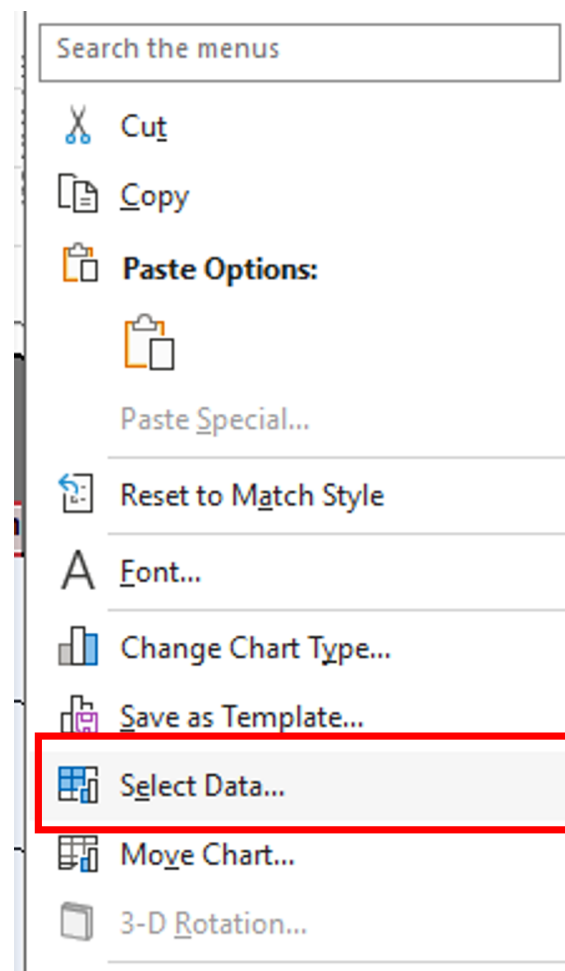


Figure A.15: Press 'Select Data..' to edit the data of a graph in Excel.

Under 'Legend Entries (Series)' press the 'Edit'-button (see Figure A.16).

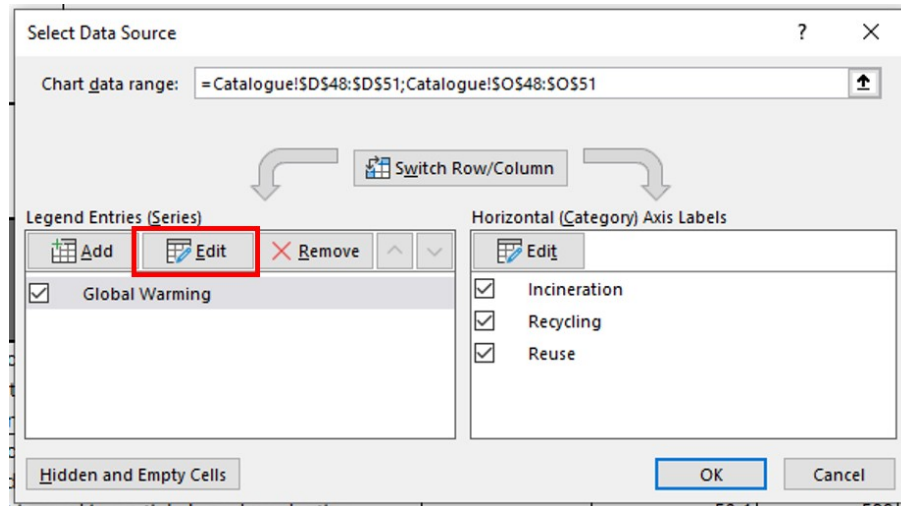


Figure A.16: Press 'Edit' under 'Legend Entries (Series)'.

The series name should refer to the drop-down cell where you choose impact category, and the series values should refer to the cells directly below that, as shown in Figure A.17.

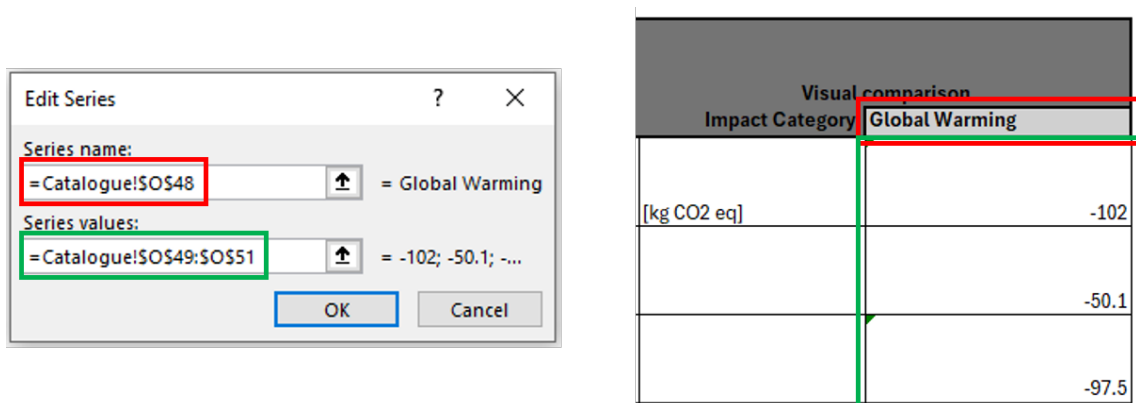


Figure A.17: Refer to the correct data in the graph.

Then, under 'Horizontal (Category) Axis Labels' press the 'Edit'-button (see Figure A.18).

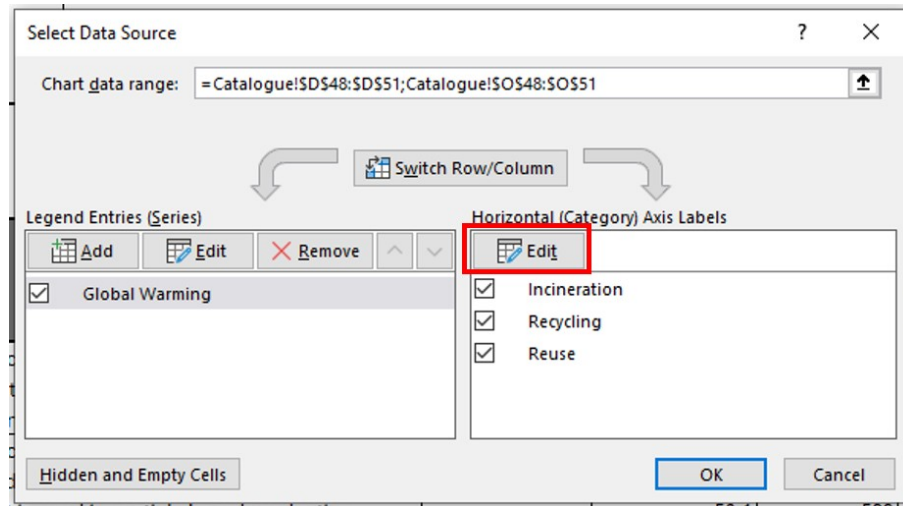


Figure A.18: Press 'Edit' under 'Horizontal (Category) Axis Labels'.

The axis label range should refer to the names of the scenarios in column D as shown in Figure A.19.

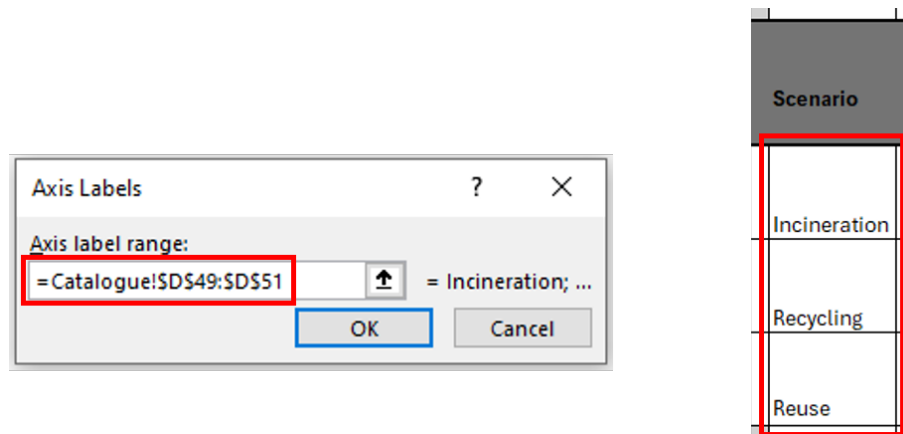


Figure A.19: Refer to the correct scenario names in the graph.

Finally, select the header of the graph and refer to the drop-down cell where you choose impact category. Once that is done, the graph should be connected to the correct data. Move it back into place and you will have successfully added a new material to the catalogue.